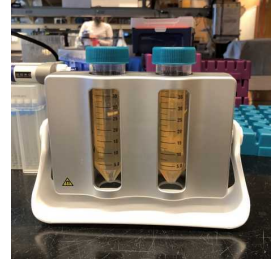


Jan 21, 2022

Manual Nanotrap Concentration and RNA Extraction for SARS-CoV-2 Viral Capture

DOI

<https://dx.doi.org/10.17504/protocols.io.b2uzqex6>



Matthew Cavallo¹, stephen.patrick.hilton¹, jvantas¹, marlene.wolfe¹, Pengbo Liu¹, Christine Moe¹, Julia Raymond¹

¹Center for Global Safe WASH, Rollins School of Public Health, Emory University, Atlanta GA USA



Stephen P Hilton

Emory University

OPEN ACCESS



DOI: <https://dx.doi.org/10.17504/protocols.io.b2uzqex6>

Protocol Citation: Matthew Cavallo, stephen.patrick.hilton , jvantas , marlene.wolfe , Pengbo Liu, Christine Moe, Julia Raymond 2022. Manual Nanotrap Concentration and RNA Extraction for SARS-CoV-2 Viral Capture. **protocols.io**

<https://dx.doi.org/10.17504/protocols.io.b2uzqex6>

License: This is an open access protocol distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working

We use this protocol and it's working

Created: December 14, 2021

Last Modified: May 18, 2026

Protocol Integer ID: 55929

Keywords: Nanotrap, manual Nanotrap, wastewater, SARS-CoV-2, COVID-19, magnetic, magnetic virus particles, Ceres, nano, rna extraction for sar, manual nanotrap concentration, rna extraction, wastewater sample, virus, extraction

Funders Acknowledgements:

Ceres Nanosciences – NIH RADx Tech

Grant ID: 75N92021C00012

Abstract

This protocol details a method for SARS-CoV-2 capture and concentration through the use of Nanotrap® Magnetic Virus Particles from 40mL of wastewater sample.

Image Attribution

Fisher Scientific

Materials

Equipment

- pH probe
- Magnetic rack (Dynamag)
- Centrifuge
- QIAamp Viral RNA Mini Kit (Cat. No. 52904 or 52906)
- Conical tube (50mL)
- Eppendorf Research Plus Single Channel Pipette
- LabGard Biological Safety Cabinet - Class 2 A2 Biosafety Cabinet
- Autoclave - Amsco Lab 240 Steam Sterilizer

Materials (per sample)

- 10 µL of BRSV
- 400 µL of  Nanotrap Magnetic Virus Particles (10) **Ceres Nano Catalog #44202**
- 1mL of  Molecular Grade Water **ATCC Catalog #60-2450**
- Three 1.7mL tubes
- 40 µL of  1X PBS (Phosphate-buffered saline)
- 560 µL of  Buffer AVL **Qiagen Catalog #19073**
- 5.6 µL of  Carrier RNA **Thermo Fisher Catalog #4382878**
- 560 µL of 96% ethanol
- 500 µL of  Buffer AW1 **Qiagen Catalog #19081**
- 500 µL of  Buffer AW2 **Qiagen Catalog #19072**
- 60 µL of  Buffer AVE **Qiagen Catalog #1020953**



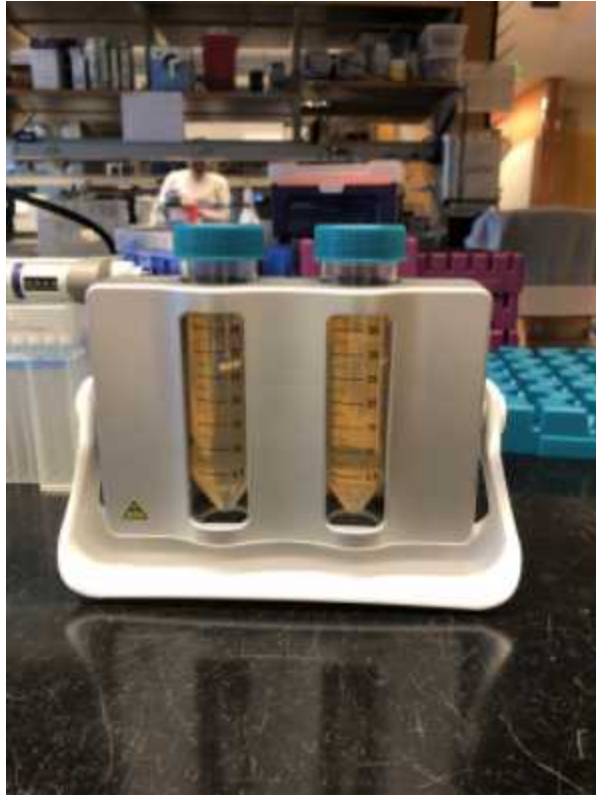
Protocol materials

- ⊗ Carrier RNA **Thermo Fisher Catalog #4382878**
- ⊗ Buffer AW1 **Qiagen Catalog #19081**
- ⊗ Buffer AW2 **Qiagen Catalog #19072**
- ⊗ Buffer AVE **Qiagen Catalog #1020953**
- ⊗ Nanotrap Magnetic Virus Particles (10) **Ceres Nano Catalog #44202**
- ⊗ Molecular Grade Water **ATCC Catalog #60-2450**
- ⊗ 1X PBS (Phosphate-buffered saline)
- ⊗ Buffer AVL **Qiagen Catalog #19073**
- ⊗ Ethanol **P212121 Catalog #BE-BDH1156**
- ⊗ double distilled water (ddH₂O)



Concentration Procedure

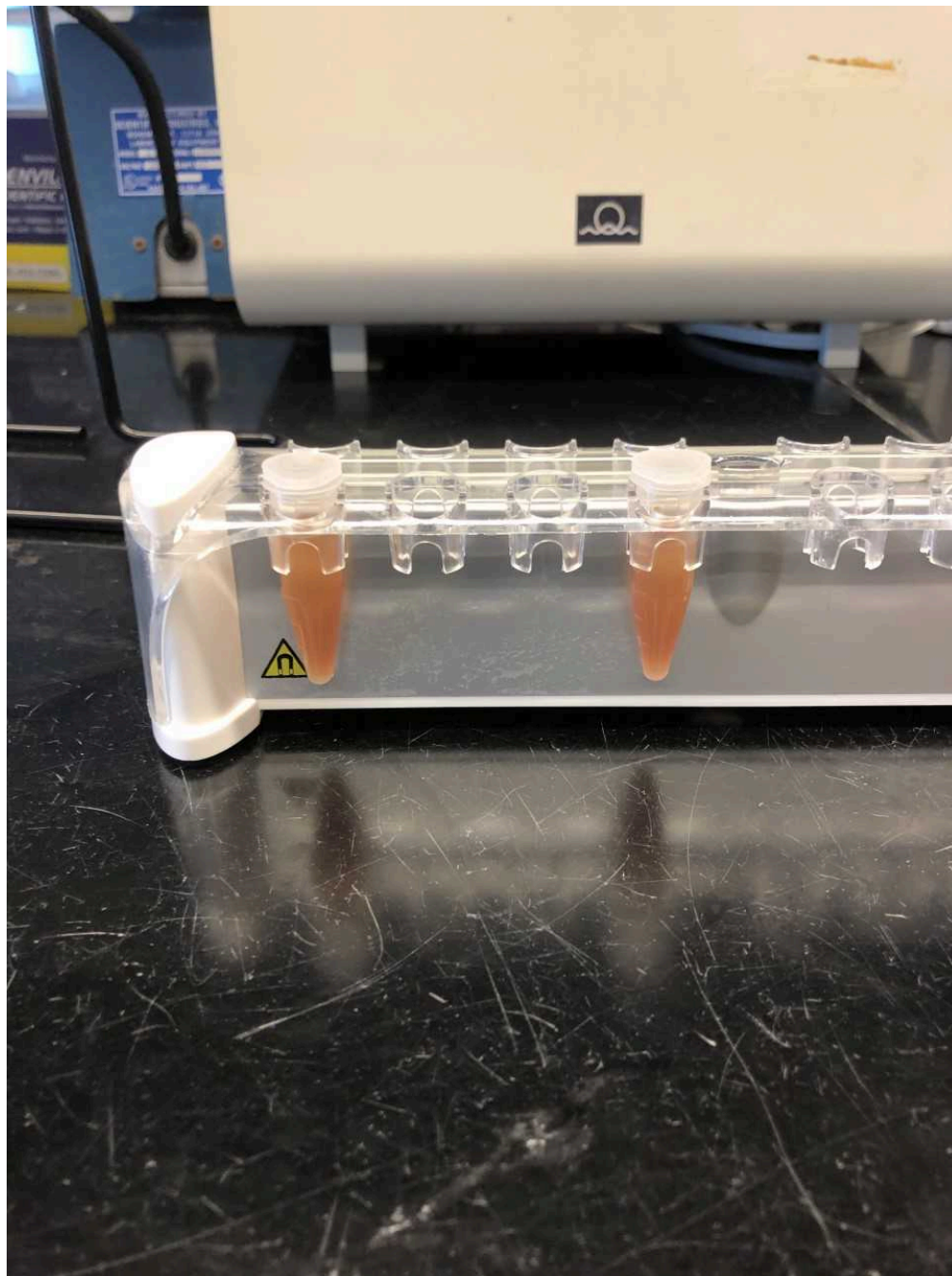
- 1 Place a 50 mL conical tube in a tube rack.
- 2 Vigorously shake the sample to re-suspend solids. Immediately move on to Step 3.
- 3 Add  50 mL of wastewater to the conical tube inside the biosafety cabinet.
- 4 Transfer  40 mL of the top supernatant into a new conical tube.
- 5 Add  10 μ L of BRSV to the conical tube.
- 6 Add  400 μ L of  Nanotrap Magnetic Virus Particles (10) **Ceres Nano Catalog #44202** to the sample and invert 2-3 times to mix together and to create a 10:1 sample volume to particle ratio.
- 7 Invert samples 3-4 times every 5 minutes at  Room temperature for  00:20:00 . 20m
- 8 Use a magnetic rack to separate the magnetic Nanotrap particles from the sample. Allow the sample to sit in the rack for at least  00:10:00 . 10m



Samples must sit in a magnetic rack for at least 10 minutes to allow for particle separation.

- 9 Use a pipette to discard the supernatant. Be careful not to disturb the red pellet of Nanotrap particles.
- 10 Add  1 mL of  Molecular Grade Water **ATCC Catalog #60-2450** to the conical tube. Vortex the tube to re-suspend the pellet.
- 11 Transfer liquid to a 1.7mL tube. Place tubes on magnetic rack and allow to sit for  00:02:00 .

2m



Sample in 1.7 mL tube sits in a magnetic rack for 2 minutes.



A red pellet of Nanotrap particles congregates to the magnetic side of the 1.7 mL tube.

- 12 Remove and discard the supernatant. Do not disturb the pellet.
- 13 Add  140 μL of 1X PBS to the particle pellet.

**Note**

See "Appendix I: Preparing Buffer Solution (1X PBS)" for instructions on how to prepare 1X PBS.

- 14 In a separate 1.7 mL tube, add  560 μL of  Buffer AVL **Qiagen Catalog #19073** and  5.6 μL of  Carrier RNA **Thermo Fisher Catalog #4382878** . Mix well by pipetting up and down several times.
- 15 In a regular, non-magnetic rack, add the Buffer AVL and carrier RNA to the particle pellet. Suspend the pellet by using the pipette to wash the sides of the conical tube with the lysis buffer, carrier RNA, and 1X PBS mixture.
- 16 Incubate the sample at  Room temperature for  00:10:00 . 10m
- 17 Use a magnetic rack to separate the magnetic Nanotrap particles from the sample. Allow sample to sit on the rack for  00:02:00 . 2m
- 18 Transfer the supernatant to a 1.7 mL tube and discard the pellet.

Extraction Step (Qiagen Viral RNA Minikit)**11m**

- 19 Add  560 μL of 96%  Ethanol **P212121 Catalog #BE-BDH1156** to sample. Mix well by pipetting up and down several times.
- 20 Add  630 μL of the sample mix onto a QIAamp Mini column.
- 21 Centrifuge sample at full speed for  00:01:00 . Once complete, discard the filtrate. 1m
- 22 Repeat Step 26 until all of the sample has gone through the column.



- 23 Transfer the QIAamp Mini Column into a new collection tube, and discard the tube containing the filtrate.
- 24 Open the QIAamp Mini Column and add  500 μL of  Buffer AW1 **Qiagen Catalog #19081** . Centrifuge the sample at full speed for  00:01:00 . Once complete, discard the filtrate. 1m
- 25 Open the QIAamp Mini Column and add  500 μL of  Buffer AW2 **Qiagen Catalog #19072** . Centrifuge the sample at full speed for  00:03:00 . Once complete, discard the filtrate. 3m
- 26 Place the QIAamp Mini Column into a new collection tube and discard the old one containing filtrate. Centrifuge the sample at full speed for  00:01:00 to remove any buffer AW2 carryover. 1m
- 27 Place the QIAamp Mini Column into a final, labeled tube and discard the old one containing the filtrate
- 28 Open the QIAamp Mini Column and add  60 μL of  Buffer AVE **Qiagen Catalog #1020953** and incubate at  Room temperature for  00:03:00 . 3m
- 29 Centrifuge the sample at  16000 rpm for  00:02:00 . 2m
- 30 Separate sample into at least two aliquots. Store in  -80 $^{\circ}\text{C}$ freezer until it is ready for PCR.

Appendix I: Preparing Buffer Solution (1X PBS)

- 31 Begin making the 10X PBS Solution by mixing the following:
-  80 g NaCl
 -  2 g KCl
 -  14.4 g Na_2HPO_4



-  2.4 g KH_2PO_4
-  800 mL Ddwater

32 Adjust the solution to  7.4 by adding 5% NaOH and/or 5% HCl.

Note

- Swirl the mixture after adding either NaOH or HCl.
- Use a pH probe in between adding NaOH or HCl to see if the pH has reached 7.4 yet.
- Wash off tip of probe with DI water in between uses.

33 Add more  double distilled water (ddH₂O) until the volume is  1000 mL .

34 Dilute the PBS solution by adding  100 mL of 10X PBS to  900 mL distilled water to create 1X PBS.

35 Dispense mixture into aliquots and autoclave for  00:20:00 . Store at room temperature.